
CloudNova AWS Deployment Project

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Introduction

- CloudNova is a portfolio website designed to showcase my skills, projects, and professional profile.
- The project involved deploying the website to the AWS cloud.
- The main focus was learning how to configure and deploy a web application using Amazon EC2.
- The deployment also involved networking and security configuration within AWS.

Business problem

- CloudNova needed a reliable way to make its website accessible online.
- Hosting the website locally would limit accessibility and availability.
- A traditional hosting solution may provide limited control over the underlying infrastructure.
- The project required a practical demonstration of cloud deployment skills.
- The solution needed to be simple, scalable, and cost-conscious.

Proposed Solution

- Deploy the CloudNova website on an Amazon EC2 instance
- Configure a Virtual Private Cloud (VPC) and public subnet for the deployment.
- Configure networking components to allow internet access.
- Configure security rules to control inbound and outbound traffic.
- Install and configure the required web server/application environment.
- Test the deployed website through its public IP address.

AWS Services Used

- **Amazon EC2** : provided the virtual server used to host the CloudNova website.
- **Amazon VPC** : provided the isolated networking environment.
- **Amazon VPC Subnet** : provided the network segment where the EC2 instance operated.
- **Internet Gateway** : enabled communication between the VPC and the internet.
- **Route Table** : controlled traffic routing from the subnet.
- **Security Group** : acted as a virtual firewall controlling access to the EC2 instance.
- **IAM** : used to manage AWS identities, permissions, and secure access.

Benefits

- **Accessibility:** The website can be accessed through the internet.
- **Flexibility:** EC2 provides control over the server environment.
- **Scalability:** EC2 resources can be adjusted as requirements grow.
- **Security:** Security Groups allow controlled network access.
- **Practical cloud experience:** Demonstrated real-world AWS deployment skills.
- **Cost awareness:** Using AWS Free Tier resources helped minimize project costs.

Challenges

- Understanding the relationship between the VPC, subnet, route table, and Internet Gateway.
- Configuring the EC2 instance correctly for web access.
- Understanding and configuring Security Group inbound rules.
- Troubleshooting connectivity and deployment issues.
- Working within AWS Free Tier limitations.
- Ensuring AWS resources were terminated after completing the project to avoid unnecessary charges.

Lessons Learned

- How cloud infrastructure components work together.
- How to launch and configure an EC2 instance.
- How AWS networking enables resources to communicate with the internet.
- How Security Groups protect cloud resources.
- How to troubleshoot common deployment and connectivity issues.
- The importance of monitoring AWS resources and managing costs.
- Cloud deployment requires both technical configuration and security awareness.

Conclusion

- The CloudNova project successfully demonstrated the deployment of a website using AWS cloud infrastructure.
- Amazon EC2 provided the computing environment required to host the website.
- AWS networking and security services enabled controlled and reliable access.
- The project strengthened my practical understanding of cloud computing, AWS infrastructure, networking, and deployment.
- This experience provides a foundation for working with more advanced AWS architectures in future projects.